PROJECT REPORT

CREDIT ASSISTANT

AI-Powered Credit Health & Financial Literacy for Indian Borrowers

Submitted for

AI in Banking, Financial Services and Insurance (BFSI) Project

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Project Repository:

https://github.com/SanakTekawade/credit-assistant

Abstract

Credit Assistant is a full-stack web application that helps Indian borrowers understand the indicators behind their credit health and learn practical steps to improve them. The application combines a financial dashboard, a credit-score view with historical trend, a what-if score simulator, a debts-and-loans tracker, an AI advisor, and a staged improvement roadmap. All monetary values are shown in Indian Rupees (₹) and the score scale follows the 300–900 range used by Indian credit bureaus.

The screenshots in this report use a demonstration profile (*Rahul Sharma*, a salaried senior software engineer based in Bengaluru) with a credit score of 698, card utilization of 36.0%, a DTI ratio of 18.8%, monthly EMIs of ₹15,000, outstanding debt of ₹4,50,000 and one missed payment in the past 12 months. These are sample values for demonstration, not verified financial records. The project is a financial-literacy and decision-support tool; it is not a credit bureau and does not make lending decisions.

1. Introduction

Access to credit depends on repayment history, credit utilization, outstanding debt, credit-history length and recent enquiries. Many borrowers find these terms hard to interpret and do not know which action would help most. Credit Assistant brings these indicators together in one place, visualises them, and turns them into plain-language, prioritised guidance.

The project belongs to the AI in BFSI domain: it pairs deterministic financial calculations with AI-generated educational recommendations. It supports awareness and planning and does not replace a regulated financial adviser, a bureau report or a lender’s underwriting.

2. Problem Statement

Borrowers often lack a single view that connects income, EMIs, card balances and credit score, and they cannot easily test how a decision (paying down a card, taking a loan) might change their score. Without that view it is hard to track progress or decide what to do first. Credit Assistant addresses this with a dashboard, a simulator, an AI advisor and a step-by-step roadmap.

3. Objectives

- Present core credit-health indicators (score, utilization, DTI, EMI, debt, missed payments) on one dashboard.
- Show the score history over selectable time ranges (1M, 3M, 6M, 1Y, All).
- Let users simulate financial actions and preview the projected score before acting.
- Track debts and loans and calculate EMI load and debt-to-income ratio.
- Provide AI-assisted, prioritised educational recommendations.
- Guide users through a staged improvement roadmap with progress tracking.
- Support secure sign-in and keep each user’s data private.

4. Scope and Key Features

4.1 Dashboard

After sign-in the dashboard greets the user and shows six summary cards: Credit Score, Card Utilization, DTI Ratio, Monthly EMI, Outstanding Debt and Missed Payments. Below them, a gauge shows the score with its band and change versus the previous score (+6 pts in the sample), and a donut chart shows revolving credit utilization with used credit (₹72,000), available credit (₹1,28,000) and the combined limit (₹2,00,000). Quick actions open the Score Simulator, Log Score and AI Advisor.

4.2 Credit Score Intelligence

This page shows a larger score gauge with the score bands (300–579 Poor, 580–669 Fair, 670–739 Good, 740–799 Very Good, 800–900 Excellent), a Credit Score History line chart with time-range filters, and a "Key Factors Impacting Your Credit Score" section that explains the standard bureau weightings: Payment History 35%, Credit Utilization 30%, Credit History Length 15%, Credit Mix 10% and New Hard Inquiries 10%. Users can log a new score or open the simulator from here.

4.3 What-If Score Simulator

The simulator lets users experiment with decisions and see the projected score. The scenarios are: pay down credit-card balances, request a credit-limit increase, complete a 12-month on-time payment streak, and take a new loan (added monthly EMI). The panel shows the current score, the simulated score and the point change, and the simulated card balance, total limit and DTI for each scenario. A simulated result can be logged as a score entry.

4.4 Debts & Loans

The Debts & Loans section tracks active obligations (four active debts in the sample profile) and feeds the monthly EMI, outstanding debt and DTI figures shown on the dashboard.

4.5 AI Advisor

The AI Advisor uses the user’s current metrics to produce an educational summary and prioritised action steps. It is powered by Google Gemini through the backend, so the API key is never exposed to the browser. Output is educational guidance, not a guarantee of score improvement.

4.6 Roadmap

"Your Path to Better Credit" is a five-stage plan with an overall completion indicator (60% in the sample, Stage 2 of 5 active). Stage 1, *Understand* (Foundation & Baselining), is 100% complete with tasks such as completing the financial profile, logging a baseline score and mapping monthly living expenses. Stage 2, *Stabilize* (Cash Flow & Debt Control), is in progress at 66% and focuses on balancing debt-to-income and eliminating immediate payment risks.

4.7 Authentication

Users register and sign in with email and password. The backend issues JWT tokens, and the Docker configuration includes a JWT secret setting for token signing. Per-user data access should be verified against the deployed backend.

5. Technology and Architecture

The repository is split into a **frontend** and a **backend** folder and is containerised with Docker Compose. The screenshots show the frontend running on port 5173. The table below summarises the stack as visible from the repository layout and Docker configuration; the exact libraries should be confirmed against the source before submission.

Layer	Technology	Purpose
Frontend	React single-page app (served by an nginx container on port 5173)	Dashboard, charts, simulator, roadmap, navigation
Backend	Python API service (container on port 8000)	Calculations, authentication, data storage, AI calls
Database	SQLite (*sqlite:////./credit_assistant.db*) on a persistent Docker volume	Users, scores, debts and assessments
Authentication	JWT (secret supplied via environment variable)	Secure sign-in and per-user access
AI	Google Gemini (*GEMINI_API_KEY* environment variable)	Educational recommendations
Deployment	Docker Compose (backend + frontend services)	One-command local deployment

5.1 High-Level Workflow

- The user signs in and completes the financial profile.
- The frontend calls the backend API with the entered values.
- The backend calculates ratios, stores records and returns the current metrics.
- The dashboard and Credit Score pages visualise current and historical data.
- The simulator projects score changes for what-if actions.
- The AI Advisor turns the metrics into prioritised guidance, and the roadmap tracks progress.

6. Financial Metrics

The formulas below are checked against the sample values shown in the screenshots.

6.1 Debt-to-Income (DTI) Ratio

DTI = monthly debt payments ÷ gross monthly income × 100. The dashboard shows 18.8% with monthly EMIs of ₹15,000, which corresponds to a gross monthly income of about ₹80,000 (15,000 ÷ 80,000 × 100 = 18.75%). The simulator's "Take a New Loan" slider raises the EMI and shows the resulting simulated DTI. The interface labels DTI below 30% as Low.

6.2 Credit Utilization

Utilization = card balance ÷ total card limit × 100. The sample card balance of ₹72,000 against a combined limit of ₹2,00,000 gives 72,000 ÷ 2,00,000 × 100 = 36.0%, matching the dashboard. Available credit is ₹2,00,000 − ₹72,000 = ₹1,28,000. The interface labels 30–50% as Moderate and advises prepaying before statement generation.

6.3 Credit Score Bands

Range	Band
300–579	Poor
580–669	Fair
670–739	Good
740–799	Very Good
800–900	Excellent

6.4 Factor Weightings

Factor	Weight
Payment History	35%
Credit Utilization Ratio	30%
Credit History Length	15%
Credit Mix	10%
New Hard Inquiries	10%

6.5 Note on the Score

Unless the app is connected to an authorised credit-bureau data source, the score is entered by the user or produced by the simulator, so it should be described as user-provided or illustrative and not as a verified CIBIL or Experian score. The simulator’s projections are estimates, not bureau calculations.

7. Screenshots of the Application

The following figures are screenshots of the running application and document the implemented interface.

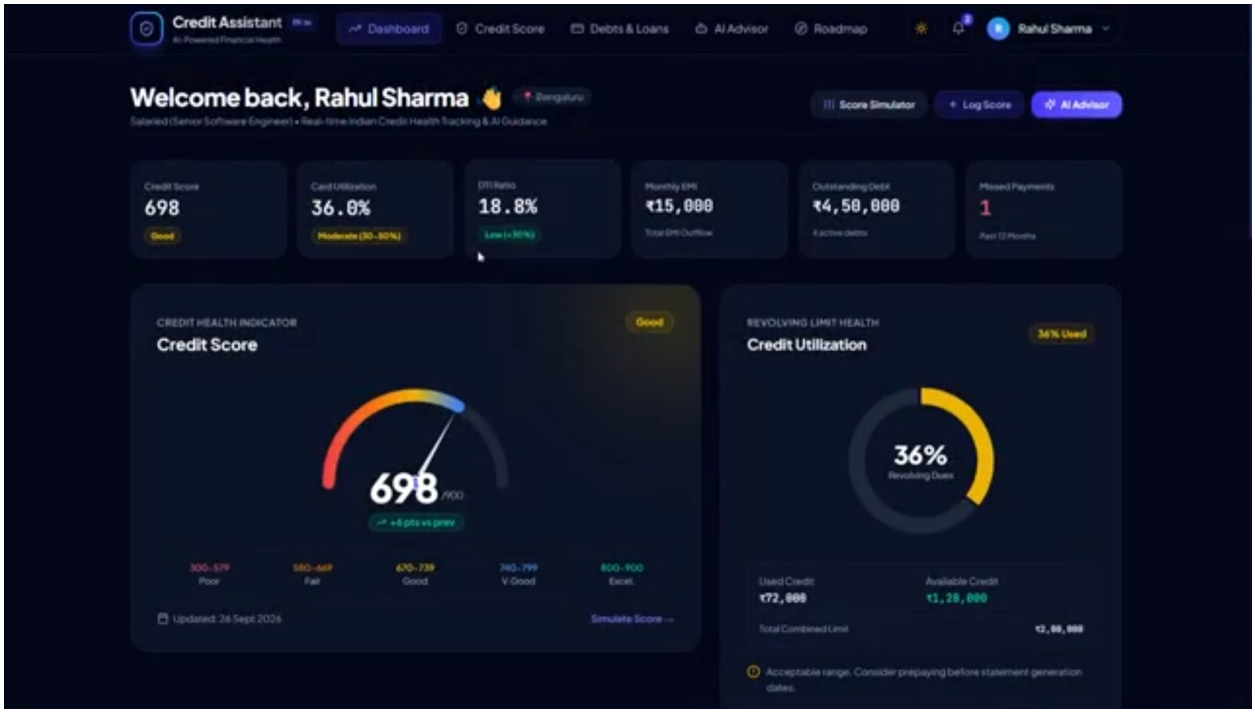


Figure 1. Dashboard showing credit score, utilization, DTI, EMI, outstanding debt and missed payments.

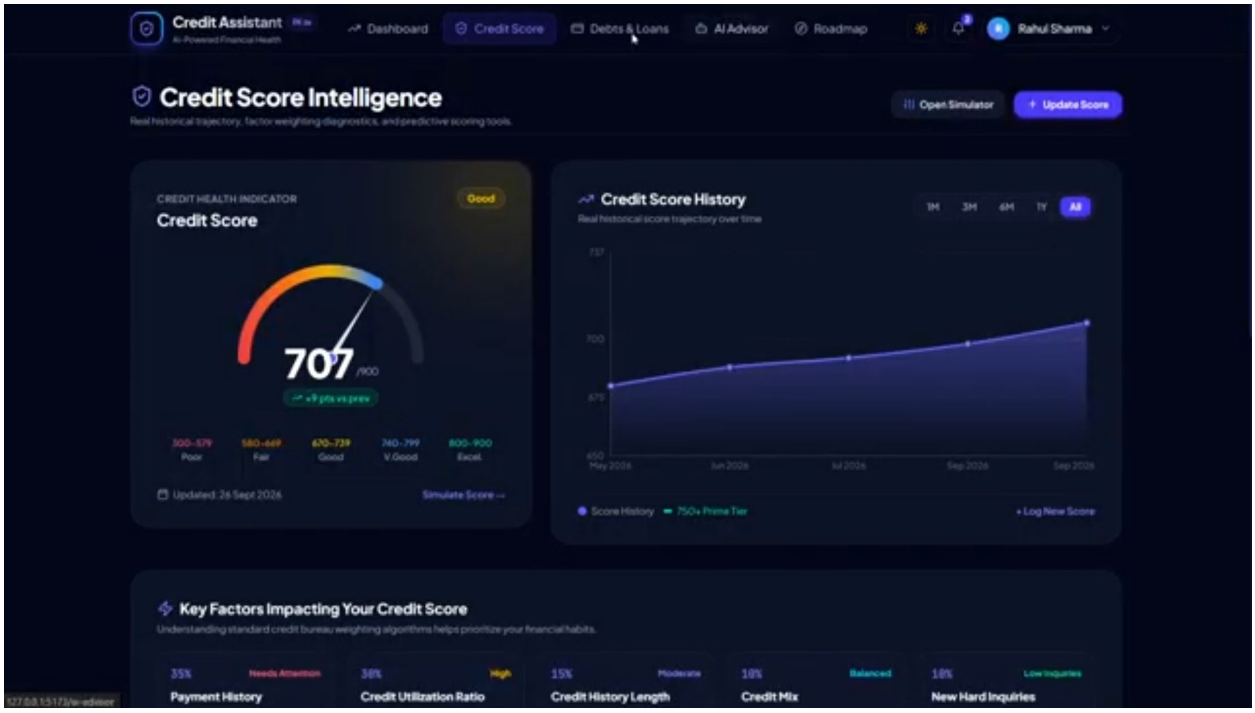


Figure 2. Credit Score Intelligence page with score gauge, history chart and key-factor weightings.

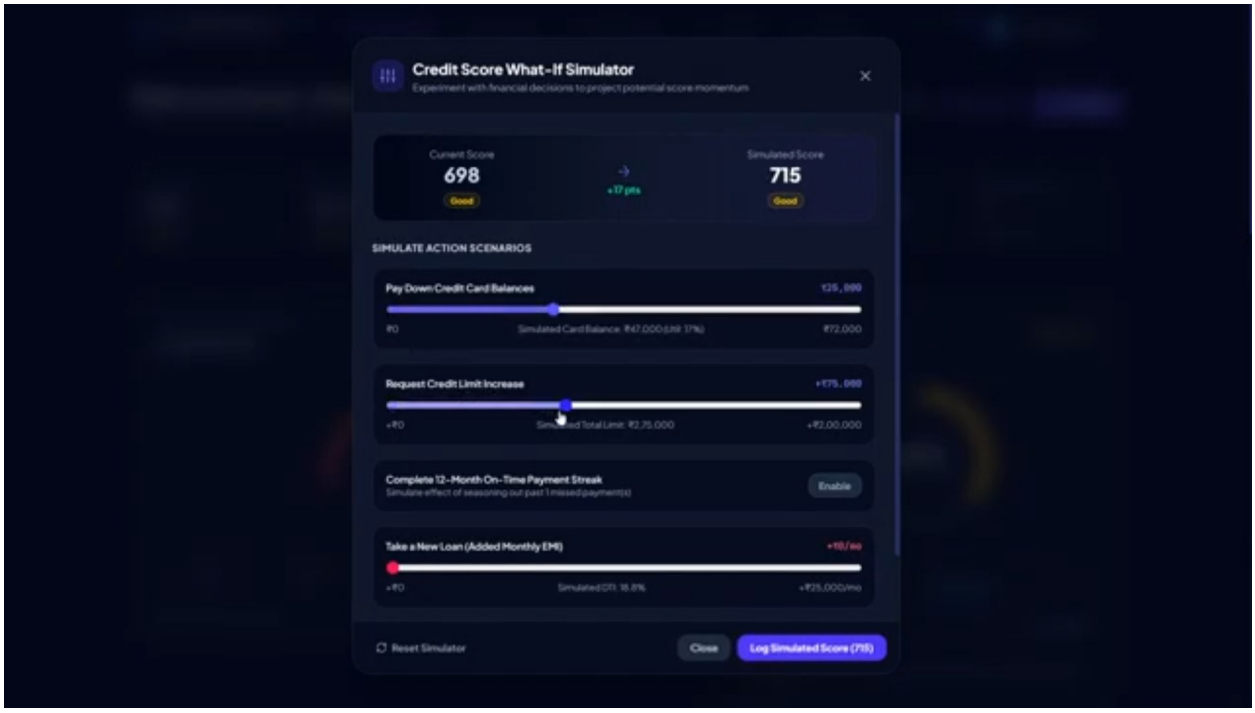


Figure 3. Credit Score What-If Simulator projecting the score from 698 to 715.

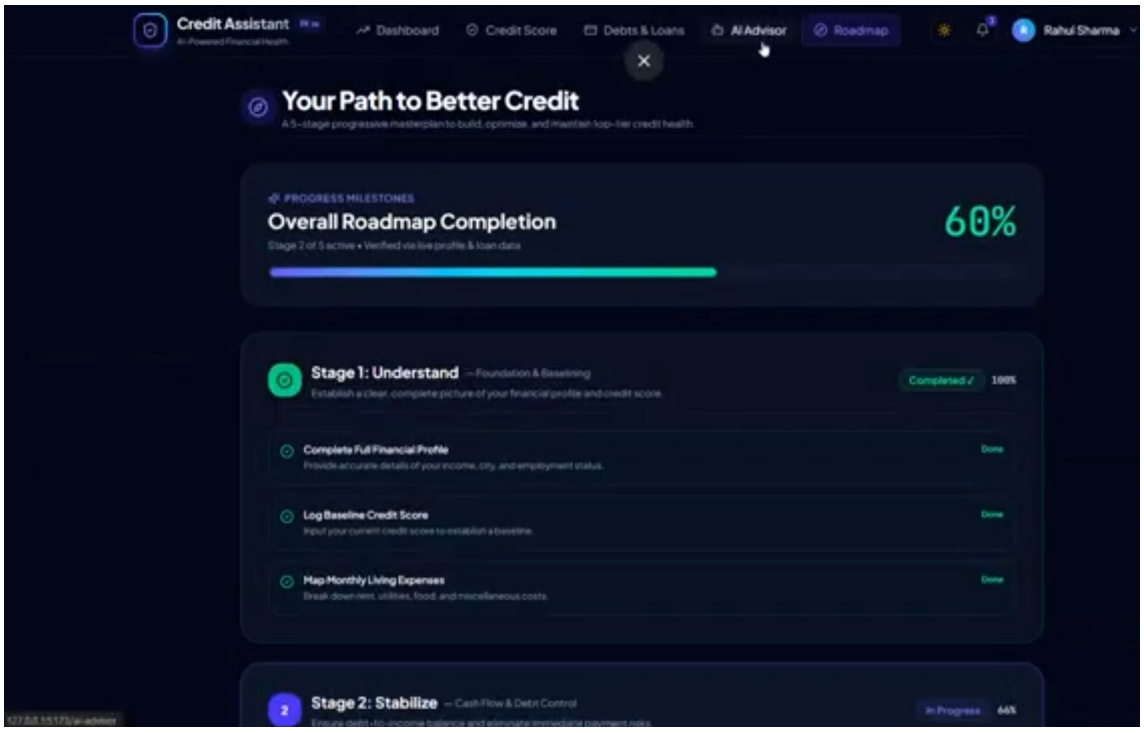


Figure 4. "Your Path to Better Credit" five-stage roadmap with overall completion tracking.

8. Security, Privacy and Responsible AI

- Use HTTPS in deployment. Keep the JWT secret and Gemini API key in environment variables or a secrets manager only.
- **Important:** the current *docker-compose.yml* contains a hard-coded JWT secret value. It is labelled as a placeholder, but it should be replaced with a private, randomly generated secret and loaded from an untracked *.env* file before any real deployment.

- Validate all numeric inputs on the backend (non-negative values, income greater than zero, card balance not above limit).
- Enforce authentication and per-user authorization on all profile, score and debt records.
- Send the AI provider only the data it needs, and avoid personally identifiable information.
- Treat AI output as untrusted text, handle API errors and rate limits, and provide a fallback when generation fails.
- Make clear that score ranges and lending thresholds are indicative and differ by bureau, lender and product.

9. Testing and Evaluation

A suitable validation plan includes unit tests for the DTI, utilization and simulator formulas; API tests for registration and login; authorization tests so one user cannot read another user's data; input-validation tests; AI-provider failure tests; and frontend checks on desktop and mobile widths. *Add the actual test results here once you have run them.*

10. Limitations

- Metrics rely on user-entered values unless a verified bureau integration is added.
- Simulator projections are estimates and may differ from actual bureau scoring.
- AI guidance can be incomplete or inaccurate and should be checked against reliable sources.
- Simple DTI or utilization thresholds cannot predict approval by every lender.
- SQLite suits a prototype; a production release would need a server database, backups, monitoring and privacy disclosures.

11. Future Enhancements

- Integrate an authorised credit-bureau data provider where legally permitted.
- Add downloadable PDF reports and exportable score history.
- Provide multilingual explanations for wider accessibility in India.
- Add EMI and payment-due reminders and a budget planner.
- Add data-source labels and calculation details beside each metric.
- Add audit logging, consent controls and a data-deletion workflow.
- Evaluate AI responses with a test set for factuality, consistency and safety.

12. Conclusion

Credit Assistant demonstrates a practical full-stack application of AI in the BFSI domain. By combining a dashboard, score intelligence, a what-if simulator, debt tracking, an AI advisor and a staged roadmap, it makes credit-health concepts easier for Indian borrowers to understand and act on. The next steps are to validate the formulas against real data, clarify the source of every score, harden security (especially secret management) and evaluate the AI recommendations before production use.

References

- Project repository: https://github.com/SanakTekawade/credit-assistant
- Google Gemini API documentation: https://ai.google.dev/gemini-api/docs
- FastAPI documentation: https://fastapi.tiangolo.com/
- React documentation: https://react.dev/
- SQLite documentation: https://www.sqlite.org/docs.html